

Feature Inventory & Roadmap

eBay Order → Google Sheets Automation Tool • Revised 22 September 2026, after the production migration

Where the project stands

The application is feature-complete and every mock, demo and simulated code path has been removed. It now runs only against real APIs: there is no placeholder connection state, no generated orders, no fixture spreadsheets, and no fallback to sample data when a credential is missing. An unconfigured deployment reports a configuration error instead.

Google is verified working against the production API — the connected account returned 100 real spreadsheets through Drive. **eBay cannot connect**: the stored Cert ID is truncated and no RuName has been created, so no order has ever been retrieved. That single blocker is what stands between this build and a working end-to-end sync.

Application code	124 files, 22,473 lines (TypeScript / React)
Automated tests	19 files, 7,705 lines — 452 passing, 0 failing
Pages	10 routes
API endpoints	27 routes
Database models	22 tables
eBay fields mappable	43 catalogued dot-paths
Transformations	20 configurable from the UI
Production build	Compiles clean

Status vocabulary: **Implemented** means the code exists, is exercised by tests, and has no mock path. **Partial** means it works but a stated limitation applies. **Blocked** means it cannot run until something outside the codebase is supplied.

1. Implemented — core pipeline

The synchronization engine is the centre of the application. One planning pass backs both the preview and the write, so what a user confirms is computed by exactly the code that then performs it.

Area	Status	Detail
eBay order retrieval	Implemented	Official Sell Fulfillment API. Paginated with cross-page de-duplication, a page guard, and a per-run API-call budget. No scraping anywhere.
Order normalisation	Implemented	Raw payload stored verbatim so a mapping added later can be re-applied. Degraded payloads absorbed without throwing; an order with no id is rejected rather than invented.
Line items and shipments	Implemented	Separate shipping-fulfillment call per shipped order, merged into the payload so <code>order.fulfillment.*</code> resolves like any other field.
Dynamic field mapping	Implemented	43 eBay fields as string dot-paths; sheet columns as rows. Adding a field or renaming a header needs no migration and does not break saved mappings.
Transformations	Implemented	20 of them: dates, currency, numbers, text cleanup, conditionals, field combining, list joining. All configurable from the UI.
Filters	Implemented	Order status, fulfillment state, marketplace, date bounds, and SKU matching in 6 modes (include / exclude / contains / starts with / ends with / all).
IF-THEN rules	Implemented	Per-row rules with ALL/ANY conditions and four actions, including switching to an alternate saved mapping set. Match counts recorded so an inert rule is visible.
Google Sheets write	Implemented	Sheets API v4. Writes only contiguous runs of mapped columns, so an unmapped column cannot be touched even accidentally.
Duplicate protection	Implemented	Keyed on eBay Order ID, or Order ID + Line Item ID in per-line-item mode. The key column is read live from the sheet, so it holds against a hand-edited sheet or a fresh database.
Row modes	Implemented	One row per order, or one row per line item. Multi-item orders summarise quantities and SKUs in order mode.

2. Implemented — safety guarantees

These are the behaviours that stop the application damaging a seller's spreadsheet. Each is enforced in code and covered by tests.

Area	Status	Detail
Only mapped columns written	Implemented	Three independent layers: the write plan, a planner refusal, and A1 ranges built from contiguous runs. Verified per-cell in tests, not just by value.
Never overwrite a used row	Implemented	A used-range probe across every known column establishes where data ends. Appends land strictly below existing content, including rows whose key cell is blank.
Refuse ambiguous sheets	Implemented	If rows hold order data but the key column is empty for all of them, the sync is blocked rather than appending a duplicate of every row.
Never delete or clear	Implemented	No call to <code>values.clear</code> , <code>batchUpdate</code> deletion, or row removal exists. The only structural change possible is appending empty rows at the bottom.

Area	Status	Detail
Controlled first sync	Implemented	The first bulk write into a destination requires an explicit preview and confirmation. Six date ranges including a custom window.
Idempotent retries	Implemented	A crash after writing but before recording does not duplicate: the next run reconciles against the sheet itself.
One sync at a time	Implemented	A SyncLock table keyed by workspace acts as a mutex, acquired before a job is claimed. A crashed holder is released by heartbeat expiry.

3. Implemented — operations and observability

Area	Status	Detail
Background scheduler	Implemented	Server-side. Runs via a long-lived worker or a platform cron hitting the tick endpoint; both are safe together. Continues with the browser closed.
Incremental sync	Implemented	Stores the last successful cutoff and syncs from there with a deliberate overlap. A single-order retry does not move the cursor.
Retry and backoff	Implemented	Exponential backoff honouring Retry-After. Only transient categories retry; a configuration error is never retried forever.
Sync history	Implemented	Sync ID, start, completion, duration, orders found, inserted, updated, skipped, failed, status. Four display statuses derived from real counters.
Per-order history	Implemented	One record per order per run: outcome, operation, sheet row, reason, error code, attempt number. An order that fails to build is recorded, not lost in a log line.
Single-order retry	Implemented	Re-runs one order through the same engine, not a shortcut path. Marks the earlier failure resolved rather than deleting it.
Raw payload viewer	Implemented	Collapsible JSON tree with filter and copy. Buyer data masked, credential-shaped fields redacted before the payload leaves the server.
Health endpoint	Implemented	Protected. Reports database, eBay and Google configuration, worker, scheduler and readiness as booleans, counts and timestamps only — never a secret value.
Startup readiness check	Implemented	Prints blockers and warnings at boot. Missing production configuration fails loudly instead of switching to anything.

4. Implemented — security and data protection

Area	Status	Detail
Token encryption	Implemented	AES-256-GCM at rest, random IV, auth tag, versioned format. Production refuses to encrypt with a weak secret rather than storing tokens poorly.
Secrets never client-side	Implemented	No NEXT_PUBLIC_ variables and no process.env in any of the 28 client components. Tokens appear only in the OAuth callbacks and the disconnect path.
CSRF protection	Implemented	Edge middleware requires a matching Origin on every mutating request. Reads are untouched; server-to-server callers without an Origin are unaffected.
Rate limiting	Implemented	Outbound pacing per connection for both APIs, plus an inbound per-IP limiter (tighter on endpoints that spend quota). eBay's deletion endpoint is exempt.
Input validation	Implemented	Zod on every request body, returning 422 with field-level errors.
Database access	Implemented	Every query scoped by user id; every mutation follows an ownership-proving lookup. No raw SQL anywhere — Prisma parameterises throughout.
Log hygiene	Implemented	A redacting logger strips credentials from messages, stacks and causes. Buyer data masked; tokens redacted even when a viewer asks to reveal.

Area	Status	Detail
Data retention	Implemented	Optional windows clear the stored eBay payload (the largest concentration of buyer personal data) and old history. Off by default — deleting a seller's data on a default would be worse than keeping it.
Account-deletion endpoint	Implemented	eBay Marketplace Account Deletion: SHA-256 challenge verification and notification acknowledgement, with 29 tests. Acknowledges only — see section 5.3.

5. What is left

Ordered by what blocks use of the product. The first item is the only thing preventing a working end-to-end sync today.

5.1 eBay credentials — blocking everything

Area	Status	Detail
eBay Cert ID	Blocked	The stored value is 36 characters ending in a 4-character group. A complete Cert ID is 44 characters ending in a 12-character group. Tested live against eBay: <code>401 invalid_client</code> . Two separate keysets have now been supplied with the same truncation, so the copy step is losing the end.
eBay RuName	Blocked	Empty. eBay's authorization-code flow redirects to an opaque RuName, not a URL, so the Connect eBay button has nowhere to send the seller. Create it under User Tokens → Get a Token from eBay via Your Application.
Consequence	Blocked	No eBay API call has ever succeeded against this deployment. The Orders page, dashboard counters and sync history are therefore empty — correctly so.

5.2 Authentication — required before public exposure

Area	Status	Detail
No sign-in exists	Partial	No route issues a session, so every request would resolve to the workspace owner. In production the application now refuses with 401 and shows a configuration page rather than serving one workspace to anyone who can reach it.
Single-user escape hatch	Partial	<code>SINGLE_USER_MODE=true</code> re-enables the owner fallback. It is an explicit assertion that the host is private — localhost, a VPN, or behind a proxy that authenticates. Not a substitute for sign-in.
What to build	Partial	A login route issuing a Session row. Everything downstream already reads the user id, so no page or query needs to change.

5.3 Compliance gap in the deletion endpoint

Area	Status	Detail
Deletion is not performed	Partial	The endpoint verifies eBay's challenge and acknowledges notifications correctly, but it does not delete the closing user's data. eBay's actual requirement is deletion.
Why it was not written	Partial	Orders are stored under the account holder's workspace and there is no mapping from an eBay <code>userId</code> to a local record. Inventing a key could delete the wrong seller's orders. The handler marks where the logic belongs.

5.4 Deployment configuration to supply

The application blocks at startup on each of these. A template is committed at `.env.production.example`.

APP_URL	Public HTTPS origin. OAuth redirects derive from it, and the session cookie is only Secure when it is https.
AUTH_SECRET	32+ random characters. Derives the token-encryption key. Changing it invalidates every stored token.
EBAY_CLIENT_ID / SECRET	Production keyset. The Cert ID must be the complete 44-character value.
EBAY_RU_NAME	Required; OAuth cannot start without it.
EBAY_ENVIRONMENT	Must be PRODUCTION. Sandbox in production is now a startup blocker.
GOOGLE_CLIENT_ID / SECRET	Already configured and verified working.
GOOGLE_REDIRECT_URI	Must match Google Cloud character for character. Localhost in production is a blocker.
DATABASE_URL	Postgres unless single-instance on a persistent disk — the sync mutex assumes one database.
CRON_SECRET	Enables the tick endpoint. Without it, run the worker process instead.
HEALTH_SECRET	Optional; falls back to CRON_SECRET. The health endpoint refuses if neither is set.

Callback URLs to register

eBay RuName accepted URL	<code>https://<domain>/api/auth/ebay/callback</code>
Google redirect URI	<code>https://<domain>/api/auth/google/callback</code>
eBay deletion endpoint	<code>https://<domain>/api/ebay/marketplace-account-deletion</code>

5.5 Worth adding, not blocking

Area	Status	Detail
Notification delivery	Partial	Automation settings store a notify-on-error address and the toggles exist, but nothing sends mail. Needs an email provider.
Webhook trigger	Partial	WEBHOOK is a recognised trigger value with no endpoint behind it. eBay push notifications would remove the polling delay.
Multi-destination sync	Partial	One active sheet configuration per workspace. The schema already permits several; the UI and engine assume one.
Notification signature check	Partial	eBay signs deletion notifications with an x-ebay-signature header. The endpoint does not yet verify it, so a forged notification would be acknowledged (it causes no action today).
Postgres migration	Partial	The schema is written to port unchanged — no enums, no JSON columns. Switching the provider and running a migration is the whole job.

6. Test coverage

452 tests across 19 files, all passing. The HTTP transports for both eBay and Google are injectable, so the full pipeline runs against recorded payload shapes and an in-memory spreadsheet that records exactly which cells were written.

Area	Status	Detail
End-to-end workflow	Implemented	25 tests walking the complete ordered workflow: headers, mapping, preview, first write, bulk sync, duplicates, updates, tracking, multi-item, missing fields, failures, scheduler.
Reliability	Implemented	16 tests: eBay and Google disconnected, expired and revoked tokens, API failures, network interruption, duplicate sync, overwrite guard, missing SKU / buyer / tracking.
Sync engine	Implemented	43 tests on inserting, duplicate protection, sync modes, line-item mode, safety refusals, filters, rules, and preview accuracy.
Order history and retry	Implemented	14 tests on per-order recording and single-order retry.
Production readiness	Implemented	21 tests on readiness checks, CSRF, rate limiting, fail-closed auth, and retention.
Security	Implemented	29 tests on token encryption, PII masking, credential redaction, and log hygiene.
Integration clients	Implemented	eBay and Google: retries, rate limits, error classification, token lifecycle, pagination, header reading.

What the tests cannot show. Everything below the socket is the real application, but the socket itself is substituted. These tests prove the application handles a response correctly; they cannot prove the live credentials work. Google has been separately verified against the production API. eBay has not, and cannot be until the credentials are complete.

7. Getting to a working sync

Three steps, in order:

1. Supply the complete eBay Cert ID and a RuName

Use the copy icon in the developer portal rather than selecting the visible text, which is ellipsized. Confirm it is 44 characters before sending. Regenerate the Cert ID if it still comes out short. This can be verified in seconds.

2. Connect both accounts in a browser

Google is already connected. eBay needs its consent flow completed once, which requires a person at the keyboard.

3. Run one controlled sync

Pick a narrow date range, review the preview, confirm. The first bulk write is gated deliberately so a mistake cannot write thousands of rows.

Until step 1 is done the Orders page will stay empty. That is the application behaving correctly: it no longer has anything to show in place of real data.